

# Block 3 - Chapter 2: Regulations, Maintenance Forms, Records & Publications

Plain-English Study Guide - read this in place of slogging the whole chapter

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**AIM Block 3 - first chapter read in class (starts Mon 2026-06-15)** | Built for Matthew | Source of truth: **FAA-H-8083-30(B), Chapter 2** (the exact assigned PDF), chapter pages 2-1 to 2-46. | No quiz / no flashcards by request - summary only. Page refs like "p. 2-14" point into the chapter PDF.

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**How to use this:** This chapter is mostly *memorization of names, numbers, and "which document does what."* There is very little hands-on skill here - it is the legal/paperwork backbone of being a mechanic. I define every term the first time it shows up, give a plain analogy, then box the must-memorize items in red. Hit the red boxes hard; those are where exam questions live.

## 1. The Big Picture: Where the Rules Live (Title 14 CFR)

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**CFR = Code of Federal Regulations.** Think of it as the giant rulebook for the whole U.S. government, split into **50 numbered "Titles."** Each Title is one subject area.

**Title 14** is the one we care about: **"Aeronautics and Space."** That is where all the FAA's aircraft rules live. (p. 2-1)

**Analogy:** Title 14 is like the "Aviation" shelf in a giant library. Within that shelf are individual books called **Parts** (Part 21, Part 43, Part 91, etc.). Within each book are **sections** (like 43.9, 91.417).

### How Title 14 is laid out

- Title 14 has **five volumes**. The **first three volumes** (containing **75 active regulations / Parts**) are the **FAA**. Volume 4 = Office of the Secretary of DOT + Commercial Space. Volume 5 = NASA. (p. 2-1)
- The FAA's airworthiness rules sort into **three categories: (1) Administrative, (2) Airworthiness Certification, (3) Airworthiness Operation**. (p. 2-1)
- Old name: **"FARs" = Federal Aviation Regulations**. The FAA now says **"14 CFR part XX"** instead, because **Title 48** ("Federal Acquisitions Regulations") also abbreviates to "FAR" and caused confusion. (p. 2-1)
- **Most Part numbers and sections are ODD-numbered** on purpose - so the FAA can slot in new even-numbered rules later (you'll see this with 43.10, 43.12, 135.422 - "note the even number" = added later). (pp. 2-1, 2-16)
- **SFAR = Special Federal Aviation Regulation:** a narrow rule for one unique situation, usually with a time limit. The SFAR number is just a sequential count - it has no link to the rule it attaches to. (p. 2-1)

### MUST KNOW - Section 1

- **14 CFR = Title 14 of the Code of Federal Regulations** = where ALL FAA rules live ("Aeronautics & Space").

- **Three categories of regs:** Administrative, Airworthiness Certification, Airworthiness Operation.
- **The 3 BIG regulations that govern airworthiness: Part 21** (certification of products/parts), **Part 43** (maintenance), **Part 91** (operating rules). Memorize these three together. (p. 2-1, 2-2)
- **SFAR** = Special FAR = narrow, usually time-limited rule.

## 2. Part 3 - General Requirements (What "Airworthy" Means)

**Part 3** is short but huge: it finally **defined the word "airworthy" in a regulation**. Before this, the term was only implied (e.g., in Part 21 section 21.183 and on the airworthiness certificate). (p. 2-1)

**Airworthy** = the aircraft **(1) conforms to its type design AND (2) is in a condition for safe operation**. Both halves are required. (14 CFR 3.5; p. 2-1)

Two more Part 3 definitions:

- **Product** = an **aircraft, aircraft engine, or aircraft propeller**. (Just those three.)
- **Record** = basically **any way information is preserved** - paper, microfilm, dataplate, stamped mark, bar code, or electronic format.

**Two kinds of airworthiness** (these two phrases run through the whole chapter):

- **Original airworthiness** = getting the design/aircraft approved and certified the first time (Part 21 land).
- **Recurrent airworthiness** = keeping it airworthy day-to-day over its life (Part 43 + Part 91 land). (pp. 2-1, 2-3)

### MUST KNOW - Section 2

- **AIRWORTHY** = **conforms to type design + in condition for safe operation**. Both parts. This is the single most-tested definition in the chapter. (Part 3.5, p. 2-1)
- **Product** = **aircraft, engine, OR propeller** (only those three).
- **Original airworthiness** (Part 21) vs. **recurrent airworthiness** (Parts 43 & 91).

## 3. Regulation vs. Guidance (the most important distinction in the chapter)

**Analogy:** A **regulation** is the **LAW** - the speed limit sign. **Guidance** is the driver's-ed manual that tells you a good way to obey the law. You **MUST** obey the law. You don't have to follow the manual word-for-word, but following it keeps you legal and safe.

|                 | REGULATION (the law)                                           | GUIDANCE (advice)                                                                |
|-----------------|----------------------------------------------------------------|----------------------------------------------------------------------------------|
| <b>Examples</b> | 14 CFR Parts (21, 43, 91...),<br>Airworthiness Directives (AD) | Advisory Circulars (AC), SAIBs, FAA handbooks,<br>manufacturer Service Bulletins |
| <b>Binding?</b> |                                                                |                                                                                  |

|                    |                                                             |                                                                                                               |
|--------------------|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
|                    | <b>Mandatory</b> - you can be violated for not following it | <b>Not mandatory</b> - "acceptable" methods, NOT the only methods... <i>unless</i> it's referenced into a reg |
| <b>Key wording</b> | "No person may...", "shall", "must"                         | "acceptable to the Administrator", "recommends", "guidance"                                                   |

**The trap:** Guidance becomes mandatory if it is "**incorporated by reference**" into a regulation. Example: an **AD** is mandatory (it's a reg, Part 39) even though it's printed in the Federal Register, not in the rulebook itself. (pp. 2-24, 2-25, 2-26)

### **MUST KNOW - Section 3**

- **Regulations = mandatory (the law). Guidance = advisory (not mandatory) UNLESS referenced into a reg.**
- **AC = advisory** (not binding). **AD = mandatory** (it IS a regulation - Part 39).
- The phrase "**acceptable to the Administrator**" signals guidance/acceptable data, NOT a hard requirement.

## **4. The Three Big Regulations - Up Close**

### **Part 21 - Certification Procedures for Products and Articles**

This is the "**birth certificate**" Part - how a design gets approved. It covers **Type Certificates (TC)**, **Supplemental Type Certificates (STC)**, **production certificates**, **airworthiness certificates**, and import/export approvals. Also **PMA** (Parts Manufacturer Approval) and **TSO** (Technical Standard Order) parts. Its biggest job is **original airworthiness**. (pp. 2-3, 2-4)

- **Section 21.50** = "Instructions for Continued Airworthiness." Any **STC made after 1981** must come with a full set of **ICA (Instructions for Continued Airworthiness)** - the maintenance/inspection instructions for the altered part. ICA is built from a **16-subject checklist**. (pp. 2-3, 2-9)

### **Part 43 - Maintenance, Preventive Maintenance, Rebuilding, and Alteration**

**This is the heart of your job.** It is the standard for maintaining every civil aircraft registered in the U.S. It has **13 rules and 6 appendices**. (p. 2-8) (Full detail in Section 7 below.)

### **Part 91 - General Operating and Flight Rules**

This is an **operating** Part aimed at the owner/operator/pilot, but the mechanic must know it because **Subpart E** (sections 91.401-91.421) is all about maintenance, preventive maintenance, and alterations. Section **91.403(b)** ties Part 91 directly to Part 43: no one may do maintenance except as Part 43 requires. **91.7(a)**: "No person may operate a civil aircraft unless it is in an airworthy condition." (pp. 2-10, 2-21)

**Quick map of the other Parts you'll see named** (don't memorize all - just recognize them): Part 1 = Definitions; Part 23 = small airplanes ( $\leq 12,500$  lb, or commuter  $\leq 19,000$  lb /  $\leq 19$  seats); Part 25 = large/transport airplanes ( $> 12,500$  lb); Part 27 = small rotorcraft ( $\leq 7,000$  lb); Part 29 = large rotorcraft ( $> 7,000$  lb); Part 33 = engines; Part 35 = propellers; Part 39 = ADs; Part 45 = ID/registration marking; Part 47 = aircraft registration; Part 65 = mechanics & other airmen; Part 121/125/135 = air carriers/operators; Part 145 = repair stations; Part 147 = AMT schools; Part 183 = FAA designees. (pp. 2-3 to 2-19)

### MUST KNOW - Section 4

- **Part 21** = certification (TC, STC, PMA, TSO) - mostly ORIGINAL airworthiness. **21.50** requires ICA.
- **Part 43** = maintenance - "the heart of aviation maintenance"; **13 rules + 6 appendices**.
- **Part 91** = operating rules; **Subpart E (91.401-91.421)** is the maintenance subpart; **91.403(b)** forces compliance with Part 43; **91.7(a)** = no operating unless airworthy.
- **ICA = Instructions for Continued Airworthiness** (16-subject checklist); required on STCs made after 1981.
- **Weight cutoffs:** Part 23 small airplane = 12,500 lb or less; Part 25 large = over 12,500 lb; Part 27 rotorcraft = 7,000 lb or less; Part 29 = over 7,000 lb.

## 5. Civil Air Regulations (CAR) - the Old Rulebook

Before today's 14 CFR existed, aircraft were certified under the **CARs (Civil Air Regulations)**, the rules of the 1940s-60s. Old aircraft are still **legally certified under the CAR they were built to**, so a mechanic may still need a CAR as the reference for older airplanes. (pp. 2-23, 2-24)

- **CAR 3** - "Airplane Airworthiness - Normal, Utility, Aerobatic, and Restricted Purpose Categories." It is the **ancestor of today's Part 23**. Examples of CAR 3 aircraft: Piper PA-22/28/32; Cessna 182, 195, 310. (p. 2-23)
- **CAR 4a** - "Airplane Airworthiness" (originated 1936, last amended Dec 15, 1952). It and CAR 4b became the regs for **large aircraft**. So: **CAR 3 = small aircraft, CAR 4a/4b = large aircraft**. (p. 2-23)

**Watch the trap:** "CAR" has **two meanings** - Civil Air Regulations (old U.S.) AND Canadian Aviation Regulations. Know which one a question means. (p. 2-23) Also: a **CAM (Civil Aeronautics Manual)** contained BOTH regulation and advisory material in one document. (p. 2-2)

### MUST KNOW - Section 5

- **CAR 3 -> Part 23 (small aircraft). CAR 4a / 4b -> large aircraft** (CAR 4b is the ancestor of Part 25).
- Old aircraft are still certified to the CAR they were built under - it can still be the maintenance reference.
- **"CAR" is a double meaning:** Civil Air Regulations OR Canadian Aviation Regulations.

## 6. Suspected Unapproved Parts (SUP)

There are **four kinds of parts** (memorize this list - it's a classic): (p. 2-24)

1. **Good part, good paperwork** -> approved, install it.
2. **Good part, BAD paperwork** -> **SUP** (suspected unapproved).
3. **BAD part, "good" (bogus) paperwork** -> **SUP** (suspected unapproved).

4. **Bad part, bad paperwork** -> unapproved - NEVER install.

The **middle two** are the **Suspected Unapproved Parts (SUP)** - if EITHER the part OR its paperwork is questionable, it's a SUP.

**What to do with a SUP:** tell the shop foreman / supervisor / quality person; **segregate and quarantine** the part until proper disposition is decided; contacting the manufacturer helps gather facts. Reference: **AC 21-29**, "Detecting and Reporting Suspected Unapproved Parts." Report a SUP at [www.faa.gov](http://www.faa.gov). (p. 2-24)

**MUST KNOW - Section 6**

- **4 part types**; the **middle two (good part/bad paperwork & bad part/bogus paperwork) = SUPs.**
- **SUP action: SEGREGATE and QUARANTINE**, notify supervisor/quality, then determine disposition.
- Reference doc = **AC 21-29.**

## 7. Inside Part 43 - The Maintenance Rules That Run Your Career

Here is what each Part 43 section actually does, in plain English. (pp. 2-14 to 2-20)

| Section | What it does (plain English)                                                                                                                                                                                                                                                                                                                    |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 43.1    | Applicability. Part 43 does NOT apply to an aircraft that has only ever held an <b>experimental</b> certificate (pink Form 8130-7). But if a Standard-cert aircraft is temporarily put into experimental, Part 43 still applies. (p. 2-14)                                                                                                      |
| 43.2    | Defines " <b>overhauled</b> " vs " <b>rebuilt</b> ." Both require: disassembled, cleaned, inspected, repaired as necessary, reassembled, tested. Difference = how it's tested: <b>OVERHAUL</b> = tested to manufacturer's <b>service limits</b> (used-part limits); <b>REBUILT</b> = tested to <b>same tolerances as a NEW part</b> . (p. 2-14) |
| 43.3    | <b>Who MAY PERFORM</b> maintenance - <b>9 persons</b> (mechanic, repairman, person under supervision, repair station, air carrier, pilot doing preventive maint. on own aircraft, etc.). (pp. 2-14, 2-15)                                                                                                                                       |
| 43.5    | Approval for <b>Return to Service (RTS)</b> - requires a proper record entry per 43.9 or 43.11; may need Form 337 for a major repair/alteration. (p. 2-15)                                                                                                                                                                                      |
| 43.7    | <b>Who MAY APPROVE</b> for RTS - <b>7 persons</b> . Note: a <b>repairman may perform work but may NOT approve RTS</b> . (p. 2-16)                                                                                                                                                                                                               |
| 43.9    | Content/form of <b>maintenance</b> record entries (NOT inspections). The <b>What / When / Who</b> rule. (pp. 2-16, 2-17)                                                                                                                                                                                                                        |
| 43.10   | Disposition of <b>life-limited parts</b> (note even number - added 2002). Defines "life-limited part" & "life status"; 7 control methods. (p. 2-16)                                                                                                                                                                                             |
| 43.11   | Content/form of <b>inspection</b> record entries. Also What/When/Who. If item <b>FAILS</b> , a <b>list of discrepancies must be given to the owner</b> . (p. 2-17)                                                                                                                                                                              |
| 43.12   | <b>Falsification</b> of records = grounds for suspension/revocation of certificate. "My signature lasts a lifetime." (p. 2-17)                                                                                                                                                                                                                  |
| 43.13   | <b>Performance Rules</b> - use the methods in the manufacturer's manual / ICA, or other methods acceptable to the Administrator. Tie-in to AC 43.13-1 / -2. (pp. 2-17, 2-18)                                                                                                                                                                    |
| 43.15   | Extra inspection rules - rotorcraft items, annual/100-hour (checklist required), progressive. Defines " <b>routine</b> " (visual, no disassembly) vs " <b>detailed</b> " (thorough, with disassembly). (p. 2-18)                                                                                                                                |
| 43.16   | <b>Airworthiness Limitations</b> - must follow the Airworthiness Limitations section of the manual/ ICA. (p. 2-18)                                                                                                                                                                                                                              |
| 43.17   | Maintenance on U.S. products by certain <b>Canadian</b> persons (AME/AMO) under the U.S.-Canada agreement. (p. 2-18)                                                                                                                                                                                                                            |

## Repair vs. Alteration, and Major vs. Minor (the decision that drives your paperwork)

Before any job, ask three questions: **(1)** What am I going to do? **(2)** How does the FAA classify it? **(3)** What documentation does it need? (p. 2-15)

- **Repair** = returns the aircraft to its **previous/unaltered** condition (new tires, brakes, magnetos, even a whole replacement engine that's the correct one). (p. 2-15)
- **Alteration** = **changes/modifies** the aircraft from its previous state (winglets, new avionics, an engine NOT listed on the TCDS). (p. 2-15)
- **Major** = could appreciably affect weight, balance, structural strength, performance, powerplant operation, flight characteristics, or other airworthiness qualities (or not done by elementary operations). **Minor** = everything else. (p. 2-15; Part 43 Appendix A)

| Job type                | Data required                                                       | Paperwork                     |
|-------------------------|---------------------------------------------------------------------|-------------------------------|
| Minor repair/alteration | <b>ACCEPTABLE data</b> (manufacturer manual, AC 43.13-1)            | Ordinary logbook entry (43.9) |
| Major repair/alteration | <b>APPROVED data</b> (AD, STC, TCDS, DER approval, FAA-approved SB) | <b>FAA Form 337</b>           |

**Field Approval:** if there's no existing approved data, the mechanic fills the front of Form 337 through block 6 (leaving block 3 open), describes the work in block 8, and sends it to the local **FSDO** (Flight Standards District Office) for an **ASI** (Aviation Safety Inspector) to approve in block 3. (pp. 2-15, 2-16)

**Preventive maintenance - what it IS:** Per Part 43, Appendix A paragraph (c), preventive maintenance is **simple/minor preservation operations and replacement of small standard parts NOT involving complex assembly operations**. A holder of at least a **private pilot certificate** may do preventive maintenance on an aircraft they **own or operate**, as long as it is NOT operated under Part 121, 129, or 135. (A **sport pilot** may do preventive maintenance only on a light-sport aircraft they own/operate.) The pilot must still make a record entry. (pp. 2-14, 2-16, 2-20)

## Part 43 Appendices (recognize what each is)

- **Appendix A** - lists Major Alterations, Major Repairs, and Preventive Maintenance. (p. 2-19)
- **Appendix B** - Recording of Major Repairs/Alterations (the **Form 337** rules). (pp. 2-19, 2-20)
- **Appendix C - Reserved** (empty). (p. 2-20)
- **Appendix D** - Scope & Detail of **Annual and 100-hour inspections** (the list is the SAME for both; only who-can-sign differs). **8 detailed areas**. (p. 2-20)
- **Appendix E - Altimeter system test** ("the 411 test") - every 24 calendar months for IFR. (p. 2-20)
- **Appendix F - ATC transponder test** ("the 413 test") - every 24 calendar months. (p. 2-20)

### MUST KNOW - Section 7 (Part 43)

- **43.3** = who may **PERFORM** (9 persons). **43.7** = who may **APPROVE** for RTS (7 persons). A repairman may **PERFORM** but **NOT** approve RTS.
- **43.9** = maintenance entries; **43.11** = inspection entries. Both = **What (work done) / When (date) / Who (signature, certificate number, type of certificate)**.
- **OVERHAUL** = tested to service limits; **REBUILT** = tested to **NEW**-part tolerances.

- **Minor = acceptable data + logbook entry. Major = approved data + FAA Form 337.**
- **Repair** returns to previous condition; **Alteration** changes it. **Major** = appreciably affects weight/balance/strength/performance/etc.
- **411 test (Appendix E) = altimeter, every 24 months. 413 test (Appendix F) = transponder, every 24 months.** Appendix C = reserved. Annual and 100-hour use the SAME Appendix D list.



## 8. Inside Part 91, Subpart E - Inspections & the Record-Keeping Rules

| Section | What it does                                                                                                                                                                                                                                                                                     |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 91.403  | <b>Owner/operator is primarily responsible</b> for keeping the aircraft airworthy, including AD compliance. (p. 2-21)                                                                                                                                                                            |
| 91.405  | Owner must get required inspections done, fix discrepancies per Part 43, and ensure entries are made; placard inoperative equipment. (p. 2-21)                                                                                                                                                   |
| 91.407  | Operation after maintenance - needs RTS + entry; an <b>operational flight check</b> is required if the work could have appreciably changed flight characteristics. (p. 2-21)                                                                                                                     |
| 91.409  | <b>Inspections. Annual</b> inspection required; if used for hire, also <b>100-hour</b> . A 100-hour may be exceeded by up to <b>10 hours</b> only to fly to the inspection facility (and that overage is subtracted from the next interval). Covers progressive inspection too. (pp. 2-21, 2-22) |
| 91.411  | "411 test" - altimeter / static system / altitude reporting, every <b>24 calendar months</b> . (p. 2-22)                                                                                                                                                                                         |
| 91.413  | "413 test" - ATC transponder, every <b>24 calendar months</b> . (p. 2-22)                                                                                                                                                                                                                        |
| 91.417  | <b>MAINTENANCE RECORDS</b> - the temporary vs permanent split (Section 9 below). (pp. 2-22, 2-23)                                                                                                                                                                                                |
| 91.419  | <b>Transfer of records</b> - records go with the aircraft when sold (may be plain language or coded). (p. 2-23)                                                                                                                                                                                  |
| 91.421  | <b>Rebuilt engine</b> records - may get "ZERO TIME" (a new record, no previous operating history) when rebuilt by the manufacturer or an approved agency. (p. 2-23)                                                                                                                              |

**Annual vs. 100-hour - the key idea:** the **inspection itself is the same** (same Appendix D list). The difference is **who can sign**: an **annual must be performed by an IA (Inspection Authorization holder)**; a 100-hour can be signed by an A&P. (Per 14 CFR 65.95(a)(2).) (pp. 2-20, 2-22)

## 9. Record-Keeping: TEMPORARY vs PERMANENT (91.417) - the most-tested split

This is the single biggest exam topic in the chapter. **14 CFR 91.417 = how long records are kept**. (Compare: **43.9 / 43.11** = how to *make* the entry; **91.417** = how long to *keep* it.) (pp. 2-22, 2-23, 2-39, 2-40)

|               | TEMPORARY records<br>91.417(a)(1) & (b)(1)                                                                                    | PERMANENT records<br>91.417(a)(2) & (b)(2)                                                                                                                                                                                                                                                                                                                |
|---------------|-------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| What they are | Records of <b>maintenance, preventive maintenance, alterations, and ALL inspections</b> - the routine "what/when/who" entries | The big-history records: <b>total time in service</b> (airframe, each engine, each propeller, each rotor); <b>current status of life-limited parts</b> ; <b>time since last overhaul</b> of items required to be overhauled; <b>current inspection status</b> ; <b>current status of applicable ADs</b> ; <b>copies of Form 337</b> for major alterations |
| How long kept | <b>1 year</b> from when the work was done - <b>OR until the work</b>                                                          | Kept <b>as long as you operate the aircraft</b> , and <b>TRANSFERRED WITH THE AIRCRAFT</b> when it is sold                                                                                                                                                                                                                                                |

|             |                                                                                                   |                                                                                    |
|-------------|---------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
|             | is repeated or superseded, whichever comes first                                                  |                                                                                    |
| Memory hook | "Temporary = the everyday entries that <b>get replaced</b> next time" (1 year / until superseded) | "Permanent = the aircraft's <b>life story</b> that <b>travels with it</b> forever" |

**Two more 91.417 pieces (don't miss these):**

- **91.417(a)** (the 411 & 413 testing records): kept **2 years** (because those tests are every 24 months). *(p. 2-22, Fig. 2-10)*
- **91.417(d)**: the **Form 337 for extended-range fuel tanks** installed in the cabin/cargo area must be **kept ON BOARD the aircraft at all times**. *(pp. 2-22, 2-20)*
- **91.417(b)(3)**: the list of **defects/discrepancies** given to the owner (per 43.11) is kept until those defects are repaired and the aircraft is approved for RTS. *(Fig. 2-10, p. 2-22)*

**Where records live:** Other than the 91.417(d) fuel-tank Form 337, maintenance records do NOT have to be carried on the aircraft - it's actually safer to keep them in an office and back them up (scan/copy) off-site against fire/flood. Records must be made available to the FAA or NTSB on request (91.417(c)). *(pp. 2-22, 2-23)*

**MUST KNOW - Section 9 (the 91.417 split - memorize cold)**

- **TEMPORARY = 91.417(a)(1) & (b)(1)** = maintenance, preventive maint., alterations, & all inspections = kept **1 YEAR or until the work is repeated/superseded**.
- **PERMANENT = 91.417(a)(2) & (b)(2)** = total time in service, life-limited parts status, time since overhaul, current inspection status, current AD status, Form 337s = **kept while you own it and TRANSFERRED WITH THE AIRCRAFT**.
- **91.417(a)** (411/413 test records) = kept **2 years**. **91.417(d)** = fuel-tank Form 337 stays **ON the aircraft**.
- **43.9 / 43.11 = how to make the entry; 91.417 = how long to keep it**.
- **Rebuilt engine -> "ZERO TIME"** record (91.421).

## 10. The FAA Document Types (what each one is FOR)

| Document                       | Full name                                  | What it's FOR / key fact                                                                                                                                                                                                                                                                                               |
|--------------------------------|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>AC</b>                      | Advisory Circular                          | <b>GUIDANCE, not binding</b> (unless referenced into a reg). Shows an "acceptable" way to comply. Numbered to match the reg subject (e.g., 43.13 relates to Part 43.13). AC system started <b>1962</b> . (pp. 2-24, 2-25)                                                                                              |
| <b>AD</b>                      | Airworthiness Directive                    | <b>MANDATORY</b> (it IS a reg - Part 39). Issued for an <b>unsafe condition likely to exist in other products of the same design</b> . Printed in the Federal Register, "incorporated by reference" into Part 39. You may operate the product <b>ONLY</b> after the AD is complied with. (pp. 2-8, 2-26)               |
| <b>SAIB</b>                    | Special Airworthiness Information Bulletin | <b>Information only - NOT mandatory</b> . Alerts/educates about a concern that does NOT rise to the level of an AD. (p. 2-27)                                                                                                                                                                                          |
| <b>TCDS</b>                    | Type Certificate Data Sheet                | The aircraft/engine/prop's " <b>spec sheet</b> " - lists limits & conditions for type certification (airspeed, weight, thrust, engine model, fuel type, min crew, etc.). It is <b>PART OF the TC</b> (per 21.41) and is evidence the product was type certificated. First published <b>Jan 1958</b> . (pp. 2-30, 2-31) |
| <b>STC</b>                     | Supplemental Type Certificate              | FAA approval of a <b>modification</b> to a type-certificated product. <b>Supplements</b> the original TC; defines the design change, how it affects type design, and serial-number effectivity. The STC & its data are the <b>property of the STC holder</b> - you must get permission to use it. (pp. 2-27, 2-30)     |
| <b>TC</b>                      | Type Certificate                           | The FAA's approval of an aircraft/engine/prop <b>design</b> (its "birth certificate"). May only be changed by FAA authorization (that change = an STC). (pp. 2-27, 2-30)                                                                                                                                               |
| <b>Aircraft Specifications</b> | (older recordkeeping document)             | FAA recordkeeping documents (from the 1926 Air Commerce Act era) for older products. <b>No longer issued</b> but still in effect. Can be converted to a TCDS at the TC holder's option. Subdivided into <b>5 groups</b> . A specification is <b>NOT part of a TC</b> . (p. 2-27)                                       |

### How AC and AD numbers are built (recognize the pattern)

- **AC number** (e.g., 25-42-C): first part = the **FAR subject/Part** (25); middle = **sequential** number within that subject (42nd); last letter = **revision** (C = 3rd revision; first issue has no letter). (p. 2-26)
- **AD number** (3 parts): **(1) year** of issue, **(2) the biweekly period** of the year, **(3) sequential** number within that biweekly period. (p. 2-26)

### Three types of AD

- **NPRM** (Notice of Proposed Rulemaking) followed by a Final Rule - the **standard** process (public comment first). (p. 2-26)
- **Final Rule; Request for Comments** - urgent enough to adopt immediately (comment period < 60 days), comments taken after. (p. 2-26)

- **Emergency AD** - immediate action needed; sent by fax/letter; effective by "**actual notice**" only to those who receive it; a matching Final Rule AD is published in the Federal Register (usually within 30 days). (p. 2-26)

**AMOC = Alternative Method of Compliance:** a different technique (or compliance time) than the AD specifies, allowed AFTER FAA approval if it gives an equal level of safety. (p. 2-27)

### Non-FAA documents

- **ATA iSpec 2200** (formerly ATA 100) - the airline-industry system that gives **standard chapter numbers** to aircraft systems so technical data is consistent across large aircraft (e.g., Ch 28 = Fuel, Ch 32 = Landing Gear, Ch 24 = Electrical Power). (pp. 2-30, 2-38)
- **Service Bulletin (SB)** - the manufacturer's notice about a defect, service change, or improvement. **NOT mandatory** unless called out by an AD or required by the inspection program. (Ref 39.27.) (p. 2-33)
- **Structural Repair Manual (SRM)** - detailed structural repair limits/criteria; usually a separate manual on large aircraft, often folded into the AMM on small ones. (p. 2-33)
- **Airworthiness Limitations** - a separate, clearly titled section of the ICA listing **mandatory replacement times, structural inspection intervals, and related procedures**. (p. 2-33)
- Manufacturer manuals: **AFM** (Airplane Flight Manual - serial-number specific) / **POH** (Pilot's Operating Handbook - model specific); **AMM** (Aircraft Maintenance Manual - whole aircraft) / **CMM** (Component Maintenance Manual - one component, often the overhaul manual). (pp. 2-33, 2-34)

### MUST KNOW - Section 10

- **AC = advisory (not mandatory). AD = mandatory (Part 39, unsafe condition likely in same-design products).**
- **SAIB = information only, not mandatory** (concern below AD level). **SB = not mandatory** unless an AD or inspection program requires it.
- **TCDS = the spec sheet; it IS part of the TC. STC = approval of a MODIFICATION, supplements the TC. TC = design approval.**
- **3 AD types:** NPRM->Final Rule (standard); Final Rule w/ Request for Comments; **Emergency AD** ("actual notice").
- **AMOC = Alternative Method of Compliance** (needs FAA approval).
- **ATA iSpec 2200 = standard chapter numbers for aircraft systems.**

## 11. The FORMS (and the Two Certificates That Live in the Aircraft)

| Form / Certificate               | What it is & key facts                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Airworthiness Certificate</b> | Shows the aircraft is airworthy. Two kinds: <b>STANDARD = FAA Form 8100-2</b> (Normal, Utility, Acrobatic, Commuter, Transport, Manned free balloon, Special classes); <b>SPECIAL = FAA Form 8130-7</b> (Primary, Restricted, Multiple, Limited, Light-sport, Experimental, Special flight permit, Provisional - the <b>experimental cert is PINK</b> ). Must be <b>ON BOARD and DISPLAYED at the cabin/flight-deck entrance, legible to passengers/crew</b> (91.203). (pp. 2-34, 2-35)                                                                                                                                                                                                                                      |
| <b>Aircraft Registration</b>     | Shows <b>ownership</b> . Applied for on <b>FAA Form 8050-1</b> ; approval issues <b>AC Form 8050-3, Certificate of Aircraft Registration</b> . A copy of the application is temporary authority to operate for up to <b>90 days</b> . Either the pink application copy or the actual certificate must be on board (91.203(a)(2)). (pp. 2-37, 2-38)                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Radio Station License</b>     | An <b>FCC</b> (not FAA) document. Required <b>ONLY if the aircraft flies OUTSIDE the U.S.</b> Not required for domestic-only operation (eliminated for domestic use in 1996). (p. 2-38)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>FAA Form 337</b>              | <b>Major Repair and Alteration</b> form. Required whenever a <b>major repair or major alteration</b> is done (the one exception: a Part 145 repair station may use a signed work order + maintenance release instead). Completed at least <b>in duplicate - original to the owner</b> , and a copy goes to the FAA. (Since fall 2005, the technician sends the copy <b>directly to the Aircraft Registration Branch, Oklahoma City</b> ; historically a copy went to the local FSDO within <b>48 hours</b> .) <b>Extended-range fuel tanks in cabin/cargo</b> require a <b>THIRD</b> copy kept <b>on board the aircraft</b> . Item 8 (back side) = description of work + the approved-data reference. (pp. 2-20, 2-38, 2-39) |

### MUST KNOW - Section 11 (forms - memorize the form numbers)

- **Standard Airworthiness Cert = FAA Form 8100-2. Special Airworthiness Cert = FAA Form 8130-7** (experimental = PINK).
- **Registration applied for on Form 8050-1; certificate issued = AC Form 8050-3.** Application = 90 days temporary authority.
- **FAA Form 337 = Major Repair and Alteration.** Original to owner; copy to FAA (now Oklahoma City Registration Branch directly). Extra fuel-tank copy stays on board.
- **Radio Station License = FCC, required only for flights OUTSIDE the U.S.**
- Airworthiness certificate must be **on board AND displayed** at the cabin/flight-deck entrance.

## 12. Records: Making Entries & Electronic Records

- **43.9 / 43.11** say you **MUST** make a maintenance/inspection entry; **91.417** says how long to keep it (see Section 9). (p. 2-39)
- **Electronic records** are allowed and now common; programs track ADs, SBs, STCs, TCDs, times in service, AD compliance, etc. The "record" definition in Part 3 already includes electronic format. The technician should know the benefits/limits of these systems. (pp. 2-1, 2-39, 2-40)

## 13. Light Sport Aircraft (LSA) Maintenance - the Basics

The **LSA** category (added in 2004) includes gliders, airplanes, gyroplanes, powered parachutes, weight-shift, and lighter-than-air. Two types: **SLSA = Special (FACTORY-built)** and **ELSA = Experimental (KIT-built)**. (p. 2-40)

**How LSA maintenance is different from regular aircraft maintenance** (this is the testable contrast):

- LSA rules come from an **industry consensus standard (ASTM)**, not the usual FAA manuals. The relevant practice is **ASTM F2483**. (pp. 2-40, 2-41, 2-44)
- **FAA Form 337 is NOT used** for LSA major repairs/alterations. Instead the work is evaluated against the ASTM standard and a **written affidavit** is provided that the LSA still meets the ASTM requirements. (p. 2-45)
- **Safety directives** (not ADs) are issued for LSA by the **manufacturer**, not the FAA. They are **mandatory - EXCEPT for experimental-use (ELSA) aircraft**. (But if the LSA contains an FAA-TC'd product, the manufacturer must issue a safety directive.) (pp. 2-45, 2-46)
- **Who can work on an LSA** (by maintenance level): (pp. 2-44, 2-45)
  - **Line maintenance** (servicing, simple R&R, the 100-hour/annual condition inspection) - minimum cert = **LSA Repairman with INSPECTION rating** (may inspect an aircraft they own).
  - **Heavy maintenance** - a certified mechanic (A and/or P) OR an **LSA Repairman with MAINTENANCE rating** who has extra task-specific training (this rating can do maintenance AND the 100-hour/annual).
  - **Overhaul** - only the manufacturer (or someone they authorize); requires a separate overhaul manual.
- **Annual condition inspection** = a detailed inspection done once a year per the LSA's maintenance manual. The **100-hour inspection is the same as the annual, just a different interval**. (pp. 2-44, 2-45)

### **MUST KNOW - Section 13 (LSA)**

- **SLSA = Special = factory-built. ELSA = Experimental = kit-built**. Category added 2004.
- **LSA uses ASTM consensus standards (F2483), NOT Form 337**. Major LSA work -> written **affidavit** of ASTM compliance instead of a 337.
- **LSA gets manufacturer "Safety Directives," not FAA ADs** - mandatory except on experimental (ELSA) aircraft.
- **LSA Repairman ratings: INSPECTION** (line maint. + 100-hr/annual on aircraft they own) vs **MAINTENANCE** (maint. + 100-hr/annual, with task-specific training). **Overhaul = manufacturer only**.

## 14. The 12 Highest-Yield Facts (your last-minute review)

If you only re-read one box before class, make it this one

1. **Airworthy** = conforms to type design + in condition for safe operation. (Part 3.5)
2. **The 3 big regs:** Part 21 (certification), Part 43 (maintenance), Part 91 (operating).
3. **AC = advisory (not binding). AD = mandatory (Part 39).** Regulation = law; guidance = advice.
4. **FAA Form 337 = Major Repair and Alteration.** Original to owner; copy to FAA (Oklahoma City).
5. **Minor = acceptable data + logbook. Major = approved data + Form 337.**
6. **43.9 = maintenance entry; 43.11 = inspection entry** (both What / When / Who).
7. **91.417 TEMPORARY (a)(1)/(b)(1):** maint., prev. maint., alterations, all inspections - kept **1 yr / until repeated or superseded.**
8. **91.417 PERMANENT (a)(2)/(b)(2):** total time in service, life-limited parts, time since overhaul, current inspection status, current AD status, Form 337s - **kept while owned & TRANSFERRED WITH THE AIRCRAFT.**
9. **OVERHAUL = service limits; REBUILT = new-part tolerances.** Rebuilt engine can get "**zero time.**"
10. **TCDS** = spec sheet, part of the TC. **STC** = approval of a modification. **SAIB** = info only.
11. **Standard A/W cert = 8100-2; Special A/W cert = 8130-7** (experimental = pink). Must be displayed in aircraft.
12. **SLSA = factory-built; ELSA = kit-built. LSA uses ASTM (not 337) and manufacturer Safety Directives (not ADs).**

Built by amt-tutor from FAA-H-8083-30(B) Chapter 2 (instructor-assigned PDF). No quiz or flashcards included by request - this is a read-through summary only. When in-class slides arrive, the worksheet/written split gets checked against AIM wording.